

Topology-to-Dimension Synthesis of a Structurally Minimal 3-DoF Closed-Chain Robotic Finger

Jaehun Kim¹, Eunho Sung¹, Seunghwan Byun¹, Seungbin You^{2,†}, and Jaeheung Park^{3,†}

Abstract—Linkage-driven robotic fingers can keep distal inertia low and avoid tendon-specific nonidealities, but achieving three-degree-of-freedom (3-DoF) motion often increases structural complexity. This paper presents a computational design framework for a linkage-driven robotic finger that achieves exactly three functional DoF with a structurally minimal closed-chain architecture. We systematically enumerate admissible 3-DoF spatial closed-chain topologies, identify the minimum-link, minimum-joint family under mobility and palm-level actuation-feasibility constraints, and then perform dimensional optimization on the selected candidate to secure sufficient fingertip workspace under symmetry and anthropomorphic constraints. The resulting mechanism places all actuators on the palm to reduce distal inertia and is realized with a geometry-based kinematic mapping that defines abduction/adduction (AA) and flexion/extension (FE) motions directly from the mechanism geometry. A fabricated prototype and a four-finger hand assembly verify that the synthesized minimal topology is not only kinematically admissible but also physically realizable as a palm-actuated robotic finger. The assembled hand performs 32 of the 33 grasp types in the GRASP taxonomy, suggesting that the reduced link-joint architecture retains sufficient functional workspace for diverse whole-hand grasping. Motion-repeatability and force-capacity tests further confirm that the realized mechanism produces consistent 3-DoF motion with practical fingertip force transmission.

Index Terms—Computational design, linkage mechanisms, closed-chain mechanisms, robotic fingers, robotic hands.

I. INTRODUCTION

THE human hand achieves versatile object manipulation through the coordinated motion of more than 20 degrees of freedom (DoF) [1]. Inspired by this biological model, many robotic hand designs aim to reproduce desired finger motions while balancing dexterity, compactness, and controllability. However, these mechanisms are typically synthesized to satisfy functional requirements rather than to formally identify a structurally minimal architecture for the required motion capability. This study systematically investigates admissible mechanism topologies for robotic fingers and seeks a design

that achieves the target motion without unnecessary structural redundancy.

Robotic hands can be broadly categorized as direct-driven, tendon-driven, or linkage-driven [2]. Direct-driven hands simplify actuation but increase distal mass and inertia [3], [4]; tendon-driven hands reduce distal inertia but suffer from friction, slack, and hysteresis [5], [6]; and linkage-driven hands avoid tendon-specific nonidealities while enabling proximal actuation, but their part count and structural complexity tend to increase with DoF [2], [7]–[11]. Among these actuation schemes, the present study focuses on linkage-driven fingers, for which greater dexterity has typically required greater structural complexity in existing designs.

Prior linkage-driven finger designs are predominantly planar and single-DoF, most commonly realized as single or cascaded four-bar linkages [7]. Although planar architectures reduce part count, they cannot realize abduction/adduction (AA) without an additional stage and thereby limit workspace and shape adaptation. To introduce AA while retaining simple flexion/extension (FE) linkages, a common strategy is to combine a planar FE module with an additional AA stage [9]. This yields greater dexterity, but the FE components are carried by the AA stage, increasing reflected rotational inertia. Moreover, adding AA to a planar FE module still does not provide full 3-DoF finger motion. To extend linkage-driven fingers to 3-DoF motion, some designs employ palm-mounted actuators and spatial kinematic chains [2], [11]. These architectures achieve compact multi-DoF motion through kinematically decoupled sub-mechanisms. However, such decoupling can increase the number of links and joints without a corresponding increase in functional DoF, resulting in greater structural complexity.

Although these designs are functionally effective, extending linkage-driven fingers from 1- or 2-DoF to 3-DoF has often been accompanied by a marked increase in structural complexity. To address this issue, this study presents a linkage-driven robotic finger that achieves exactly three functional DoF with a structurally minimal architecture. The selected topology is dimensionally optimized into a finger-like mechanism with palm-mounted actuation, and the fabricated prototype is used to validate that the synthesized minimal topology can be realized as a functional robotic finger and assembled into a multi-finger hand. The main contributions of this work are as follows:

- 1) **Topology-level synthesis:** a systematic framework that enumerates admissible 3-DoF spatial closed-chain topologies and selects a feasible finger topology with the minimum number of links and joints under mobility and actuation-feasibility constraints.

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TABLE I
ADMISSIBLE LINK-JOINT COMBINATIONS YIELDING 3-DOF.

No. of links	No. of joints	Joint type		
		1-DoF	2-DoF	3-DoF
3	3	0	0	3
4	4	0	3	1
	4	1	1	2
	5	0	0	5
⋮				

- 2) **Geometry-level optimization:** a dimensional optimization on the selected minimal topology to secure sufficient fingertip workspace within an anthropomorphic finger envelope.
- 3) **Prototype realization and functional validation:** a hardware implementation demonstrating that the synthesized minimal topology can be realized as a palm-actuated robotic finger and assembled into a multi-finger hand capable of representative 3-DoF motions and diverse grasp postures.

The remainder of the paper details the topology search, dimensional design, prototype realization, kinematic validation, and functional capability assessment.

II. MECHANISM TOPOLOGY SELECTION

Designing a linkage-driven finger begins with selecting a mechanism topology. This section enumerates admissible link-joint combinations for 3-DoF spatial closed-chain mechanisms, identifies the minimum-link, minimum-joint topologies under functional and topology-level feasibility criteria, and imposes bilateral symmetry to select the topology for dimensional optimization.

A. Enumeration of Admissible Link-Joint Combinations

To address the structural complexity of existing 3-DoF finger mechanisms, we seek a feasible 3-DoF spatial closed-chain topology with the minimum number of links and joints. To ensure a systematic and exhaustive search, we enumerate admissible link-joint combinations for spatial closed-chain mechanisms whose mobility satisfies the spatial Grübler equation [12]:

$$\text{DoF} = 6(n_l - 1) - \sum_{j=1}^{n_j} (6 - f_j), \quad (1)$$

where n_l and n_j denote the counts of links and joints, respectively, and $f_j \in \{1, 2, 3\}$ is the DoF of the j th joint. The search space ranges from a minimal 3-link mechanism to larger assemblies and includes many combinations satisfying DoF = 3. Representative admissible combinations are listed in Table I.

For compact integration, all three 1-DoF actuators must be mounted on a common ground link. At the combination level, this leads to two actuation-feasibility conditions: (i) the mechanism must contain at least three 1-DoF joints, and (ii) the

TABLE II
CANDIDATE LINK-JOINT COMBINATIONS YIELDING 3-DOF UNDER ACTUATION-FEASIBILITY CONDITIONS.

No. of links	No. of joints	Joint type		
		1-DoF	2-DoF	3-DoF
6	7	3	0	4
7	8	3	3	2
	8	4	1	3
⋮				

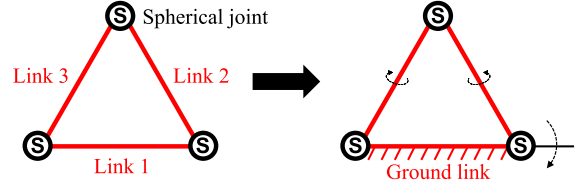


Fig. 1. Example 3-link-3S mechanism.

link-joint graph must be sufficiently connected to admit a link incident to at least three joints during topology instantiation. To conservatively reflect the latter requirement before explicit topology instantiation, we retain only combinations satisfying $n_j \geq n_l + 1$. Applying these actuation-feasibility conditions together with the 3-DoF mobility requirement yields the candidate link-joint combinations listed in Table II.

B. Topology Instantiation and Feasibility Screening

Algebraic DoF counts alone do not guarantee functionally effective finger motion. For example, the canonical 3-link-3S mechanism satisfies the spatial Grübler equation for DoF = 3, yet the axial spins of the two non-ground links do not transmit to the output, leaving only a single independent output motion, as illustrated in Fig. 1. We therefore exclude any topology in which one or more joint motions do not produce independent distal output motion. This criterion eliminates the six-link/seven-joint candidate in Table II, whose admissible instantiations contain S-S links that introduce axial spins analogous to the 3-link-3S example. Accordingly, we proceed with the seven-link/eight-joint candidates in Table II, the minimum-link, minimum-joint family remaining after this functional screening. Each admissible link-joint combination is then instantiated as an explicit link-joint topology by assigning the joints to specific links. We retain only topologies satisfying the previously introduced actuation-feasibility requirement that at least three 1-DoF joints be incident to a common ground link. This conversion yields 32 distinct seven-link/eight-joint topologies shown in Fig. 2.

Two additional topology-level feasibility issues are then considered. First, some instantiated topologies split into separate sub-mechanisms from the ground link and therefore cannot be realized as a single 3-DoF finger mechanism with one distal output link. Second, in many topologies, two of the three 1-DoF joints designated for actuation on the ground link are additionally connected by a sub-mechanism consisting of three

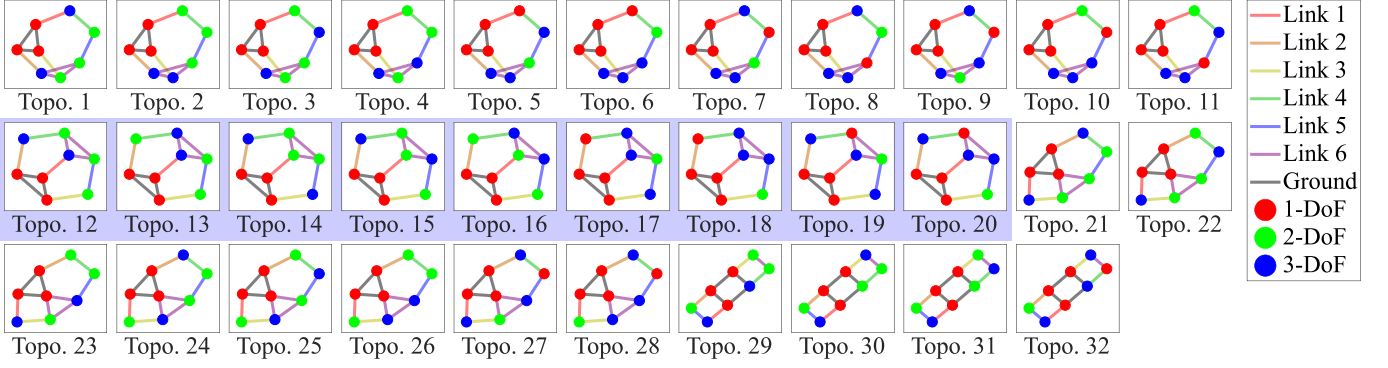


Fig. 2. Seven-link/eight-joint topologies enumerated from the admissible link–joint combinations. The 9 topologies remaining after discarding those with topology-level infeasibility issues are highlighted.

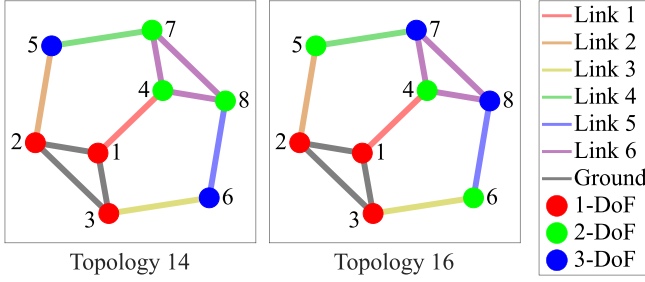


Fig. 3. Two mirror-symmetric seven-link/eight-joint topologies selected from the admissible set.

non-ground links. If actuators are placed at both of these 1-DoF joints, the resulting sub-mechanism becomes locally over-constrained and kinematically inconsistent. Thus, even when a link–joint combination is admissible at the combination level, its specific connectivity pattern can still induce partial over-constraints [13]. After discarding all topologies that exhibit these topology-level infeasibility issues, 9 admissible topologies remain (topologies 12–20 in Fig. 2).

C. Symmetry-Based Final Topology Selection

Imposing bilateral symmetry—desirable for uniform fingertip behavior—reduces the 9 admissible seven-link/eight-joint topologies to two mirror-symmetric candidates that share the same joint census (three 1-DoF, three 2-DoF, and two 3-DoF), as illustrated in Fig. 3. The two candidates differ only in the ordering of the 2-DoF and 3-DoF joints along the two spatial branches originating at the actuated 1-DoF joints. We adopt topology 14 to place higher rotational mobility closer to the base. This choice is further motivated by the expectation that allocating larger relative orientation changes to a proximal high-mobility joint can mitigate workspace loss due to passive-joint mechanical limits [14].

An instantiation of the adopted seven-link/eight-joint topology in a finger-like morphology is shown in Fig. 4. The *ground link* (gray) anchors the mechanism to the palm and houses three co-located actuators. The *end-effector link* (purple)—the only link not incident to ground—lies on the sagittal symmetry axis and is realized as a tetrahedral shell terminating at the fingertip. To keep design options open, the *remaining*

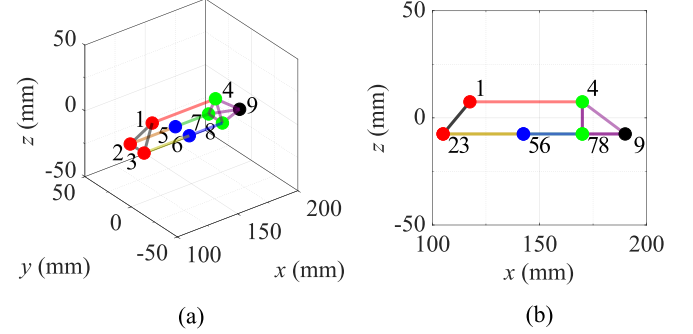


Fig. 4. Adopted seven-link/eight-joint topology instantiated in a finger-like morphology: (a) Isometric view, (b) Side view.

links are not pre-assigned to specific phalanges. This mapping preserves the kinematic integrity of the seven-link/eight-joint mechanism, satisfies palm-level packaging by grounding all actuators, and shifts mass proximally. In our implementation, the three 1-DoF joints are realized as revolute joints, the three 2-DoF joints as universal joints, and the two 3-DoF joints as spherical joints.

III. DIMENSIONAL OPTIMIZATION

After selecting an admissible, symmetric, and structurally minimal 3-DoF spatial closed-chain topology, the remaining challenge is to determine a functional finger geometry for the selected topology. Although the overall design goal is structural minimality, the selected minimal topology must still provide a functional fingertip workspace. Therefore, after fixing the link–joint census, we use workspace-based objectives as dimensional design criteria to secure usable motion within an anthropomorphic envelope. Accordingly, this section details the parameterization, design domain, objective functions, anthropomorphic constraints, and multi-objective search procedure.

A. Parameterization and Symmetry Constraints

The mechanism is parameterized by the Cartesian coordinates of eight spatial joints and the fingertip $\mathbf{p}_i = (x_i, y_i, z_i)^T \in \mathbb{R}^3$ for $i = 1, \dots, 9$ (27 parameters), and by

the directions of three unit input axes $\hat{\mathbf{u}}_1, \hat{\mathbf{u}}_2, \hat{\mathbf{u}}_3 \in \mathbb{R}^3$ (6 parameters), yielding 33 decision variables in total.

To obtain a unique optimal solution up to length scaling and rigid-body transformations, we fix the initial coordinates of two points in the global hand frame, whose origin is placed at the wrist ($x = 0$). The fixed x -coordinates are chosen from mean anthropometric measurements of male middle fingers reported in [15]:

- **Fingertip position (3 constraints):** the endpoint of the tetrahedrally shaped end-effector link is aligned with the human fingertip location and is fixed at

$$\mathbf{p}_9 = [194.1, 0, 0]^T \text{ mm}, \quad (2)$$

which corresponds to the reported wrist-to-fingertip distance of the middle finger (digit 3 height = 194.1 mm).

- **MCP-aligned input joint (3 constraints):** among the three actuated joints, the one lying on the symmetry axis is aligned with the human metacarpophalangeal (MCP) joint location and is fixed at

$$\mathbf{p}_1 = [110.3, 0, 0]^T \text{ mm}, \quad (3)$$

which is obtained from the reported wrist-to-fingertip distance minus the middle finger length (digit 3 height – digit 3 length = 194.1 – 83.8 = 110.3 mm).

For the MCP-aligned input joint, the associated input axis is also constrained for orthogonality:

- **Central input-axis orthogonality (2 constraints):** align the central input axis perpendicular to the sagittal plane $y = 0$ (i.e., parallel to $\hat{\mathbf{y}}$):

$$\hat{\mathbf{u}}_1 = [0, 1, 0]^T. \quad (4)$$

Finally, for the remaining joints, we impose mirror symmetry about the sagittal plane in the initial configuration:

- **Mid-plane anchoring (1 constraint):** anchor the remaining central joint on the sagittal plane:

$$y_4 = 0. \quad (5)$$

- **Joint-coordinate symmetry (9 constraints):** for symmetric pairs $(i, j) \in \{(2, 3), (5, 6), (7, 8)\}$, enforce reflection about the sagittal plane:

$$x_i = x_j, \quad y_i = -y_j, \quad z_i = z_j. \quad (6)$$

- **Input-axis symmetry (2 constraints):** enforce mirror directions for the AA pair of inputs:

$$u_{x,2} = -u_{x,3}, \quad u_{y,2} = u_{y,3}, \quad u_{z,2} = -u_{z,3}. \quad (7)$$

After applying the position-fixing and symmetry reductions, $33 - 20 = 13$ design variables remain for optimization.

B. Design Domain

Except for the two fixed points, the remaining seven spatial joints $\mathbf{p}_i = (x_i, y_i, z_i)^T \in \mathbb{R}^3$ for $i = 2, \dots, 8$ are restricted to lie within a common rectangular design domain that approximates the volume occupied by a male middle finger in the hand frame, based on anthropometric measurements reported in [15]. The longitudinal bound is taken from the reported

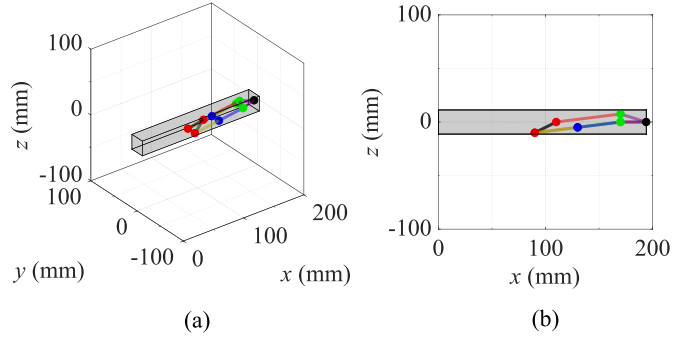


Fig. 5. Rectangular design domain that approximates the volume of a human finger: (a) Isometric view, (b) Side view.

wrist-to-fingertip distance of the middle finger (digit 3 height = 194.1 mm), so that the design domain spans from the wrist ($x = 0$) to the fingertip ($x = 194.1$). The transverse bounds are taken from the reported proximal interphalangeal joint breadth of the middle finger (digit 3 proximal interphalangeal joint breadth = 22.5 mm), using half of this value in both the y - and z -directions. Specifically, the initial joint coordinates are constrained to satisfy

$$\begin{aligned} x_i &\in [0, 194.1] \text{ mm}, \\ y_i &\in [-11.25, 11.25] \text{ mm}, \\ z_i &\in [-11.25, 11.25] \text{ mm}. \end{aligned} \quad (8)$$

Fig. 5 illustrates the resulting design domain in the hand frame.

C. Objective Functions

With the link–joint census fixed by the topology-level synthesis, workspace-based objectives are used to select dimensions that secure usable fingertip motion within an anthropomorphic envelope. In particular, we consider not only the full three-dimensional reachable set of the fingertip, but also the subset of configurations where the fingertip lies on the sagittal symmetry plane. To capture both aspects, we formulate two scalar objectives that operate on (i) a voxelized 3D workspace and (ii) a pixelized 2D workspace on the symmetry plane.

Objective 1 — Voxelized workspace volume, f_1 : Let $\mathcal{W}_{\text{mech}}$ denote the reachable fingertip workspace of the mechanism. On a uniform voxel grid $\mathcal{V}$ with voxel volume $\Delta V = (5 \text{ mm})^3$, define the occupancy indicator $\chi_{\mathcal{W}}(\mathbf{v}) \in \{0, 1\}$, which is 1 if voxel $\mathbf{v}$ is occupied in the workspace $\mathcal{W}$ and 0 otherwise. We maximize the total voxel volume of the fingertip workspace:

$$f_1 = \sum_{\mathbf{v} \in \mathcal{V}} \chi_{\mathcal{W}_{\text{mech}}}(\mathbf{v}) \Delta V. \quad (9)$$

Objective 2 — Pixelized workspace area on the symmetry plane, f_2 : Let Π_{sym} denote the sagittal symmetry plane ($y = 0$). On a uniform pixel grid $\mathcal{P} \subset \Pi_{\text{sym}}$ with pixel area $\Delta A = (5 \text{ mm})^2$, define the in-plane occupancy indicator $\chi_{\mathcal{W}}^{\Pi}(\mathbf{p}) \in \{0, 1\}$, which is 1 if pixel $\mathbf{p}$ is occupied by the intersection of $\mathcal{W}_{\text{mech}}$ with Π_{sym} . We then maximize the fingertip workspace area on the symmetry plane:

$$f_2 = \sum_{\mathbf{p} \in \mathcal{P}} \chi_{\mathcal{W}_{\text{mech}}}^{\Pi}(\mathbf{p}) \Delta A. \quad (10)$$

D. Anthropomorphic Constraints

We enforce anthropomorphic consistency via hard constraints; configurations that violate any constraint are deemed infeasible and excluded from objective evaluation. First, to prevent the fingertip from moving behind the back of the hand and to ensure that the finger remains usable for interacting with objects in front of the palm, we restrict the feasible workspace so that the fingertip never crosses the dorsal side of the hand. For any configuration η considered in the workspace computation, we require

$$z_9(\eta) \leq 0, \quad (11)$$

where $z_9(\eta)$ denotes the z -coordinate of the fingertip in the hand frame for configuration η .

Second, to avoid unrealistically folded or non-anthropomorphic shapes that increase workspace at the expense of human-like morphology, we impose ordering constraints on the initial x -coordinates of the joints along each kinematic branch connecting the ground and end-effector links. In the initial configuration, the joint coordinates must satisfy

$$\begin{aligned} x_1 &\leq x_4 \leq x_9, \\ x_2 &\leq x_5 \leq x_7 \leq x_9, \\ x_3 &\leq x_6 \leq x_8 \leq x_9 \end{aligned} \quad (12)$$

which enforce a monotonic progression from the ground link toward the fingertip along the central, left, and right branches, respectively.

Finally, to ensure that the fingertip bends toward the palm rather than hyperextending in the opposite direction, we constrain several inter-link angles in the sagittal plane. All joint angles are defined on the sagittal symmetry plane Π_{sym} by orthogonally projecting the relevant link directions onto Π_{sym} ; angles follow a right-hand rule convention about the $+y$ axis. For any configuration η , we require

$$0^\circ \leq \theta_{XZ,579}(\eta), \theta_{XZ,689}(\eta) \leq 180^\circ \quad (13)$$

where $\theta_{XZ,ijk}(\eta)$ denotes the angle, in the XZ -plane, between the segments connecting points $i-j$ and $j-k$ at configuration η . We then define the feasible design set as the collection of parameter vectors for which all of the above constraints are satisfied for all configurations η used in the workspace evaluation.

E. Multi-Objective Search with NSGA-II

Based on this formulation, we executed 8 independent NSGA-II runs and formed the union of the Pareto sets across runs [16] (each run: 100 iterations, population size 50, crossover percentage 0.7, mutation percentage 0.4, mutation rate 0.02). Initial design variables were sampled uniformly within the design domain, and only feasible candidates $\mathbf{x} \in \mathcal{X}_{\text{feas}}$ were admitted to selection and variation.

To choose a single best-compromise design, we selected the *knee point* on the union Pareto front [17] as the Pareto-optimal solution. Fig. 6(a) shows the outermost Pareto front among the 8 NSGA-II runs and highlights the selected Pareto-optimal solution. Using the corresponding design variables, Fig. 6(b)

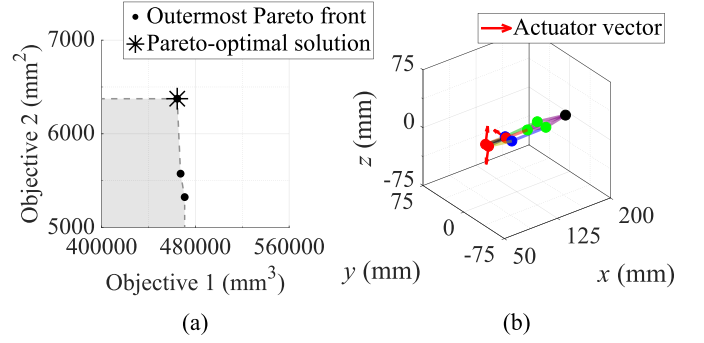


Fig. 6. (a) Pareto-optimal solution in the objective space from 8 NSGA-II runs, (b) Selected Pareto-optimal mechanism in the initial configuration.

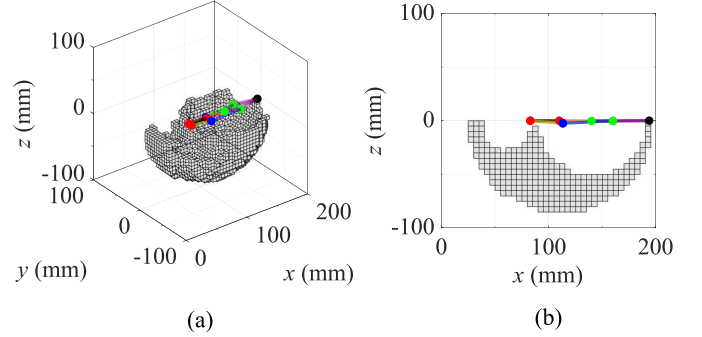


Fig. 7. Fingertip workspace of the selected Pareto-optimal mechanism: (a) Voxelized workspace in 3D, (b) Pixelized workspace on the symmetry plane.

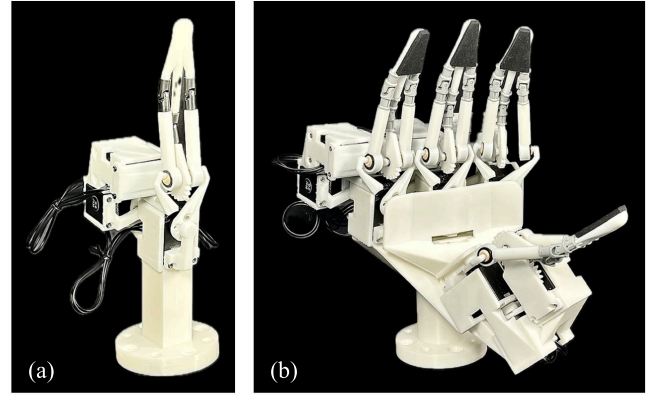


Fig. 8. (a) Fabricated finger mechanism, (b) Assembled robotic hand with four identical finger mechanisms integrated on the palm.

shows the selected mechanism in the initial configuration, while Fig. 7(a) shows the voxelized fingertip workspace for Objective 1 f_1 , and Fig. 7(b) shows the pixelized fingertip workspace on the symmetry plane for Objective 2 f_2 .

IV. PROTOTYPE REALIZATION AND KINEMATIC VALIDATION

A. Mechanical Design and Hardware Implementation

Guided by the optimization results in Sec. III, we fabricated a finger mechanism to validate geometry, clearances, and joint placement against the anthropomorphic mapping, as shown in Fig. 8(a). All links were 3D printed in resin with dimensional tolerances specified to maintain joint coaxiality and

limit assembly play. Three miniature universal joints (Miyoshi Kikai, M series, M-3.5-A-A) were employed to realize the required 2-DoF joints. For the 3-DoF spherical joints, we used commercial spherical plain bearings (igus, igubal series, KGLM-03), providing the required multi-axis articulation with compact packaging.

For actuation, we used three Dynamixel XC330-T288-T units as ground-mounted inputs at joints 1, 2, and 3, consistent with the optimization's input placement. Due to packaging constraints around joint 1 (axis proximity to neighboring links), a direct motor-on-axis mount was impractical. We therefore used a 1:1 gear pair to offset the motor from the input axis, preserving the transmission ratio while improving clearance. In addition, to keep each finger module as compact as possible in width, the motors for joints 2 and 3 were mounted coaxially on their respective input axes and positioned to minimize module width while maintaining service access.

Finally, four identical fingers were integrated onto a palm (ground link) to assemble a complete hand, as shown in Fig. 8(b). For rapid and low-cost full-hand integration, the universal joints in the hand assembly were realized using plastic components (LEGO, Technic series, Universal Joint 3L). The assembled hand, including all finger mechanisms and ground mounting, weighs approximately 600 g.

B. Kinematic Modeling and Joint-Space Mapping

Because the proposed finger uses a structurally minimal 3-DoF spatial closed-chain linkage, its motion does not lend itself to a one-to-one correspondence with canonical anatomical joint angles. We therefore define three geometry-based angles and numerically map them to actuator commands for the three ground-mounted revolute inputs.

Let the desired angles be

$$\boldsymbol{\theta}_{\text{des}} = [\theta_{\text{AA}} \quad \theta_{\text{FE,MCP}} \quad \theta_{\text{FE,IP}}]^T, \quad (14)$$

where IP denotes the interphalangeal joint, and let the actuator command vector be

$$\mathbf{q}_{\text{act}} = [q_1 \quad q_2 \quad q_3]^T. \quad (15)$$

1) *Geometric definition of the desired angles:* To define $\boldsymbol{\theta}_{\text{des}}$ using a minimal set of consistent geometric features, we use (i) the MCP-aligned joint 1, (ii) the fingertip point 9, and (iii) the tetrahedral end-effector shell whose distal face is (7, 8, 9). We regard the distal face (7, 8, 9) of the shell as a distal phalanx proxy, and the adjacent virtual face (1, 7, 8)—which shares the edge (7, 8) and includes joint 1—as a proximal phalanx proxy.

We define the vertical plane Π_{AA} as the plane passing through $\mathbf{p}_1$ and $\mathbf{p}_9$ and parallel to the global z -axis, as illustrated in Fig. 9(a). The azimuth of Π_{AA} is defined as θ_{AA} :

$$\theta_{\text{AA}} = \text{atan2}(y_9 - y_1, x_9 - x_1), \quad \theta_{\text{AA}} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right). \quad (16)$$

The unit normal to Π_{AA} is then given by

$$\hat{\mathbf{n}}_{\text{AA}} = [-\sin \theta_{\text{AA}} \quad \cos \theta_{\text{AA}} \quad 0]^T. \quad (17)$$

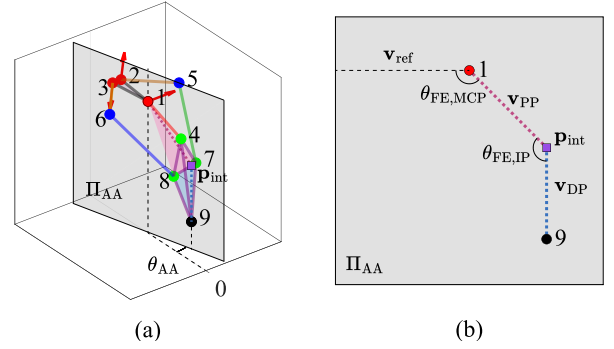


Fig. 9. Geometric construction of the task-space angles used for kinematic modeling: (a) three-dimensional construction on the finger mechanism, (b) in-plane construction on Π_{AA} .

Next, we define $\mathbf{p}_{\text{int}}$ as the intersection of the segment $\overline{\mathbf{p}_7\mathbf{p}_8}$ with Π_{AA} :

$$\mathbf{p}_{\text{int}} = \mathbf{p}_7 + t(\mathbf{p}_8 - \mathbf{p}_7), \quad \hat{\mathbf{n}}_{\text{AA}}^T(\mathbf{p}_{\text{int}} - \mathbf{p}_1) = 0, \quad (18)$$

where $t \in [0, 1]$ indicates that the intersection lies within the segment.

To define an MCP reference direction on Π_{AA} , we orthogonally project the origin $\mathbf{O} = [0, 0, 0]^T$ onto Π_{AA} :

$$\mathbf{O}_{\Pi_{\text{AA}}} = \mathbf{O} - (\hat{\mathbf{n}}_{\text{AA}}^T(\mathbf{O} - \mathbf{p}_1))\hat{\mathbf{n}}_{\text{AA}}. \quad (19)$$

We then define the in-plane directions on Π_{AA} , as illustrated in Fig. 9(b):

$$\mathbf{v}_{\text{ref}} = \mathbf{p}_1 - \mathbf{O}_{\Pi_{\text{AA}}}, \quad \mathbf{v}_{\text{PP}} = \mathbf{p}_{\text{int}} - \mathbf{p}_1, \quad \mathbf{v}_{\text{DP}} = \mathbf{p}_9 - \mathbf{p}_{\text{int}}, \quad (20)$$

where $\mathbf{v}_{\text{PP}}$ and $\mathbf{v}_{\text{DP}}$ represent the proximal and distal phalanx proxy directions within Π_{AA} , respectively. The angles $\theta_{\text{FE,MCP}}$ and $\theta_{\text{FE,IP}}$ are computed as signed angles between the corresponding proxy directions about $\hat{\mathbf{n}}_{\text{AA}}$:

$$\begin{aligned} \theta_{\text{FE,MCP}} &= \text{atan2}\left(\hat{\mathbf{n}}_{\text{AA}}^T(\mathbf{v}_{\text{ref}} \times \mathbf{v}_{\text{PP}}), \mathbf{v}_{\text{ref}}^T \mathbf{v}_{\text{PP}}\right), \\ \theta_{\text{FE,IP}} &= \text{atan2}\left(\hat{\mathbf{n}}_{\text{AA}}^T(\mathbf{v}_{\text{PP}} \times \mathbf{v}_{\text{DP}}), \mathbf{v}_{\text{PP}}^T \mathbf{v}_{\text{DP}}\right). \end{aligned} \quad (21)$$

2) *Constraint-based forward kinematics:* For a given $\mathbf{q}_{\text{act}}$, the mechanism posture is obtained by solving the closed-chain constraint equations

$$\mathbf{F}(\boldsymbol{\xi}; \mathbf{q}_{\text{act}}) = \mathbf{0}, \quad (22)$$

where $\boldsymbol{\xi}$ collects the pose parameters of the non-ground links. The constraint solver is warm-started from the previously converged state, and a tetrahedral consistency constraint for the end-effector shell is enforced. After obtaining the joint positions $\{\mathbf{p}_i(\mathbf{q}_{\text{act}})\}$, the angles $\boldsymbol{\theta}(\mathbf{q}_{\text{act}})$ are computed by (16)–(21).

3) *Numerical inverse mapping:* Given $\boldsymbol{\theta}_{\text{des}}$, we solve for $\mathbf{q}_{\text{act}}$ such that $\boldsymbol{\theta}(\mathbf{q}_{\text{act}})$ matches $\boldsymbol{\theta}_{\text{des}}$ by minimizing a three-angle residual. Because each residual evaluation requires solving (22), we use continuation in $\boldsymbol{\theta}$ with warm-starting and multi-start local search (Levenberg–Marquardt via `lsqnonlin`, otherwise trust-region via `fsolve`) to obtain a consistent solution within the feasible workspace.

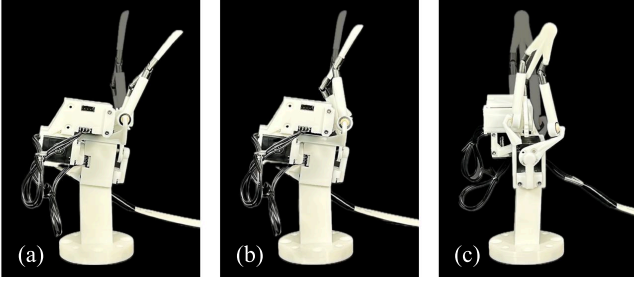


Fig. 10. Representative finger motions obtained from the proposed kinematic mapping: (a) MCP FE (side view), (b) IP FE (side view), and (c) AA (front view).

Fig. 10 demonstrates that the geometry-based mapping produces interpretable finger-level motions on the realized minimal topology. This result confirms that the non-anatomical closed-chain architecture can still be commanded in functionally meaningful finger-level coordinates.

V. FUNCTIONAL CAPABILITY ASSESSMENT

Since the proposed design prioritizes structural minimality at the topology level, the following tests examine whether the reduced link-joint architecture still provides sufficient whole-hand grasp diversity, motion consistency, and practical contact force in the physical implementation.

A. Whole-Hand Grasp Capability

To assess whole-hand pose diversity, the assembled four-finger hand was tested against the GRASP taxonomy proposed by Feix *et al.* [18]. As shown in Fig. 11, the hand performed 32 of the 33 grasp types; the only unsuccessful type was *Prismatic 4 Finger*, which requires a thumb opposing four fingers. This result indicates that the reduced link-joint architecture retains sufficient functional workspace and pose diversity for diverse grasp postures, supporting the main design objective of preserving functional capability with minimal links and joints.

B. Motion Repeatability

Motion repeatability was evaluated to verify that the reduced link-joint architecture produces consistent fingertip motion after physical implementation. Test 1–5 correspond to five predefined motions: MCP FE, IP FE, AA, simultaneous MCP FE + IP FE, and simultaneous MCP FE + IP FE + AA, respectively. Each command was repeated for 32 cycles, and the absolute deviation from the cycle-averaged fingertip displacement was measured using a dial indicator.

Fig. 12 shows the distribution of the absolute fingertip displacement error for the five tests. Across all tests, the mean error remained approximately 0.01–0.02 mm, with the largest mean error of approximately 0.03 mm in Test 2. These results indicate that the realized minimal closed-chain topology produces consistent fingertip motion while accommodating passive multi-DoF joints and a non-anatomical actuation structure.

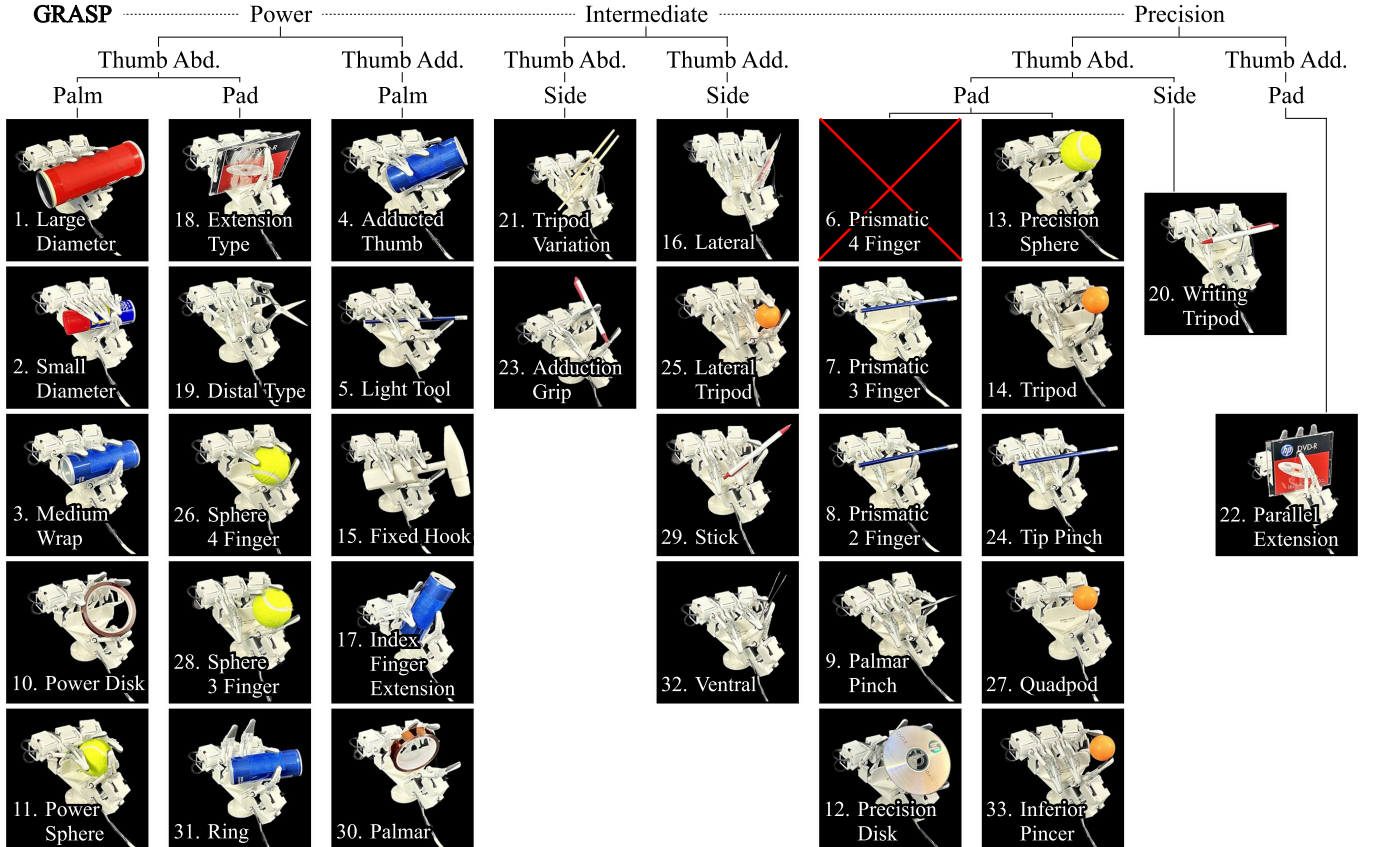


Fig. 11. Whole-hand grasp postures evaluated based on the GRASP taxonomy. The assembled four-finger robotic hand performs 32 of the 33 grasp types.

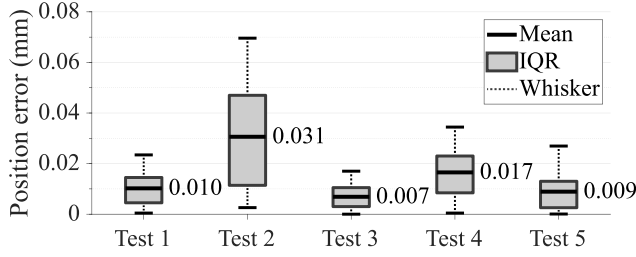


Fig. 12. Repeatability results for five predefined motion tests.

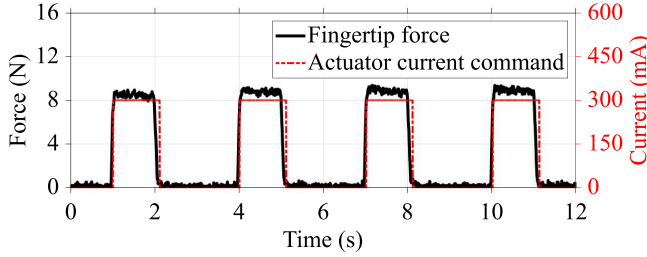


Fig. 13. Measured fingertip force capacity under a constant-current command.

C. Force Capacity

Fingertip force capacity was evaluated using a 6-axis force/torque sensor (OnRobot HEX) with the finger and sensor rigidly mounted to a common base. All three actuators were operated in current-control mode with an identical command of 300 mA, and the command was applied and released four times. The fingertip force was computed as

$$F_{\text{mag}} = \sqrt{F_x^2 + F_y^2 + F_z^2}.$$

Fig. 13 shows the measured force profiles. The average peak fingertip force was approximately 8.7 N, suggesting that the reduced link-joint architecture retains practical force-transmission capability for object interaction.

VI. CONCLUSION AND FUTURE WORK

This paper presented a computationally designed linkage-driven robotic finger that achieves exactly three functional DoF with a structurally minimal spatial closed-chain architecture. To this end, we systematically enumerated admissible 3-DoF topologies, identified the minimum-link and minimum-joint family under functional and actuation-feasibility constraints, and then optimized the selected candidate to secure sufficient fingertip workspace within an anthropomorphic design envelope.

A fabricated prototype and its geometry-based kinematic mapping demonstrated representative MCP FE, IP FE, and AA motions on the realized minimal topology. A four-finger robotic hand assembled from the same modules performed 32 of the 33 grasp types in the GRASP taxonomy, indicating that the reduced link-joint architecture retains sufficient functional workspace for diverse whole-hand grasping. Motion-repeatability and force-capacity tests further confirmed consistent fingertip motion and practical force-transmission capability in the physical implementation.

Overall, the results suggest that systematic topology-to-dimension synthesis can yield compact and functional linkage-driven finger architectures that preserve useful motion and grasp capability despite structural minimality. Nevertheless, the non-decoupled actuation space of the proposed mechanism prevents a simple one-to-one mapping from desired motions to actuator commands. Future work will leverage the same systematic design framework while introducing carefully chosen additional linkage structures to improve controllability and anthropomorphic similarity without incurring excessive structural complexity.

REFERENCES

- [1] I. A. Kapandji, "The physiology of the joints, volume I, upper limb," *American Journal of Physical Medicine & Rehabilitation*, vol. 50, no. 2, p. 96, 1971.
- [2] U. Kim, D. Jung, H. Jeong, J. Park, H.-M. Jung, J. Cheong, H. R. Choi, H. Do, and C. Park, "Integrated linkage-driven dexterous anthropomorphic robotic hand," *Nature Communications*, vol. 12, no. 1, p. 7177, 2021.
- [3] H. Liu, K. Wu, P. Meusel, N. Seitz, G. Hirzinger, M. Jin, Y. Liu, S. Fan, T. Lan, and Z. Chen, "Multisensory five-finger dexterous hand: The DLR/HIT Hand II," in *2008 IEEE/RSJ international conference on intelligent robots and systems*. IEEE, 2008, pp. 3692–3697.
- [4] D.-H. Lee, J.-H. Park, S.-W. Park, M.-H. Baeg, and J.-H. Bae, "KITECH-hand: A highly dexterous and modularized robotic hand," *IEEE/ASME Transactions on Mechatronics*, vol. 22, no. 2, pp. 876–887, 2017.
- [5] *Shadow Dexterous Hand: Technical Specification*, Shadow Robot Company, Jun. 2024, accessed: 2026-01-07. [Online]. Available: https://www.shadowrobot.com/wp-content/uploads/2024/06/20240610-UPDATED-shadow_dexterous_hand_e_technical_specification.pdf
- [6] C.-H. Xiong, W.-R. Chen, B.-Y. Sun, M.-J. Liu, S.-G. Yue, and W.-B. Chen, "Design and implementation of an anthropomorphic hand for replicating human grasping functions," *IEEE Transactions on Robotics*, vol. 32, no. 3, pp. 652–671, 2016.
- [7] E. Difonzo, G. Zappatore, G. Mantriota, and G. Reina, "Advances in finger and partial hand prosthetic mechanisms," *Robotics*, vol. 9, no. 4, p. 80, 2020.
- [8] X. Liu, X. Zheng, and S. Li, "Development of a humanoid robot hand with coupling four-bar linkage," *Advances in Mechanical Engineering*, vol. 9, no. 1, p. 1687814016686313, 2017.
- [9] E. Sung, S. You, S. Kim, and J. Park, "Snu-avatar robot hand: Dexterous robot hand with prismatic four-bar linkage for versatile daily applications," in *2023 IEEE-RAS 22nd International Conference on Humanoid Robots (Humanoids)*. IEEE, 2023, pp. 1–8.
- [10] S. Chung, E. Sohn, and S. Jeong, "A robotic finger with a 4-bar linkage-based compact and continuously variable active transmission," *IEEE Robotics and Automation Letters*, vol. 9, no. 7, pp. 6304–6311, 2024.
- [11] T. Zaw, D. Mahalingam, N. Baiata, A. Patankar, and N. Chakraborty, "Design and Kinematic Analysis of a Novel Series-Parallel Hybrid Finger for Robotic Hands," *Journal of Mechanisms and Robotics*, vol. 17, no. 4, p. 044512, 2025.
- [12] L.-W. Tsai, *Robot analysis: the mechanics of serial and parallel manipulators*. John Wiley & Sons, 1999.
- [13] J. S. Dai, Z. Huang, and H. Lipkin, "Mobility of overconstrained parallel mechanisms," *Journal of Mechanical Design*, vol. 128, no. 1, pp. 220–229, 2006.
- [14] J.-P. Merlet, "Determination of the orientation workspace of parallel manipulators," *Journal of intelligent and robotic systems*, vol. 13, no. 2, pp. 143–160, 1995.
- [15] T. M. Greiner, *Hand anthropometry of US army personnel*. US Army Natick Research, Development & Engineering Center, 1991.
- [16] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE transactions on evolutionary computation*, vol. 6, no. 2, pp. 182–197, 2002.
- [17] J. Branke, K. Deb, H. Dierolf, and M. Osswald, "Finding knees in multi-objective optimization," in *International conference on parallel problem solving from nature*. Springer, 2004, pp. 722–731.
- [18] T. Feix, J. Romero, H.-B. Schmiedmayer, A. M. Dollar, and D. Kragic, "The grasp taxonomy of human grasp types," *IEEE Transactions on human-machine systems*, vol. 46, no. 1, pp. 66–77, 2015.